

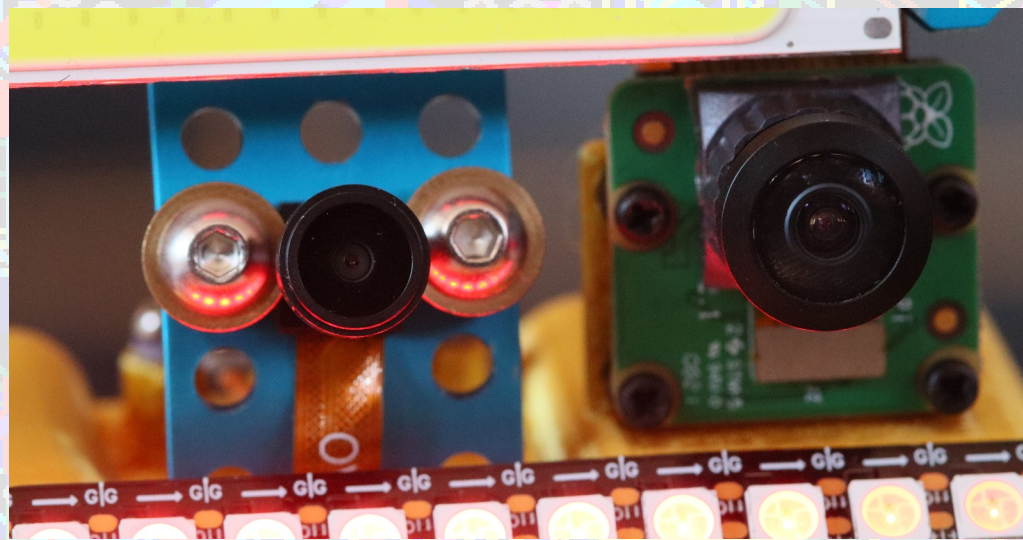


TEAM BIGG-IRMI

Rescue Line



Front Camera Module

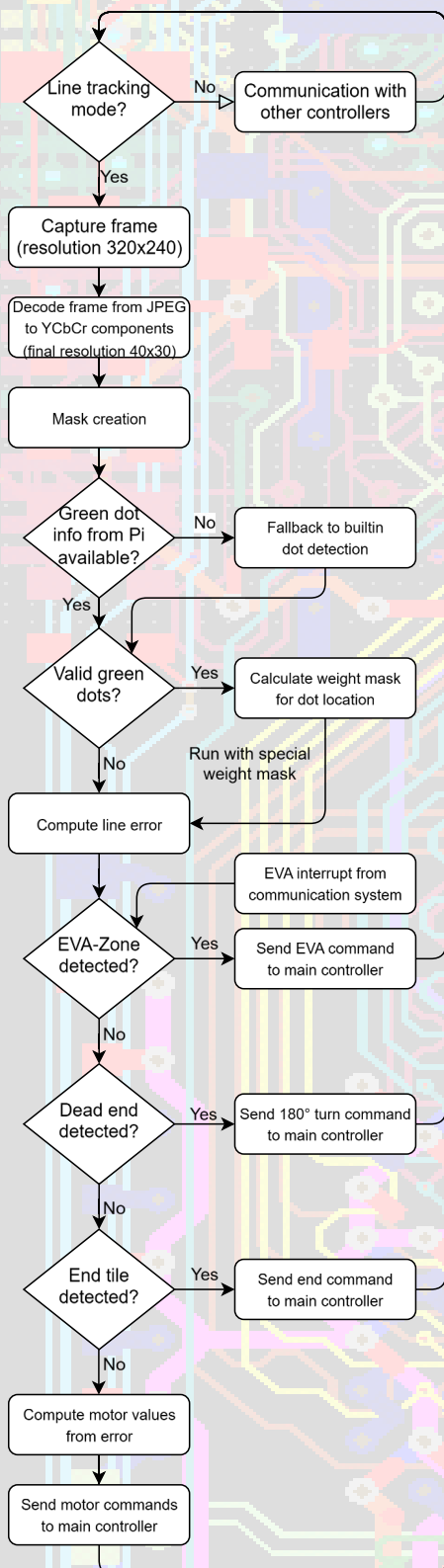


This module comprises one OV2640 camera with a 120° field of view (FOV) and one IMX708 camera with a 160° FOV, with one camera connected to an ESP32S3 microcontroller. The second camera is connected to a Raspberry Pi 5, which runs local, self-trained AI models to assist in detecting the silver tape at the entrance to the evacuation zone. The Raspberry Pi also assists the line-tracking camera in detecting green dots at intersections, as the OV2640 has a very limited color range.



Software

Our software is made up of several elements, each of which runs on a separate microcontroller or single board computer (SBC). Each element contains our custom-written HLDC byte-stuffing UART communication class, enabling it to easily send motor commands, read sensor values, and control servos. All programs are written in C++, except for the SBC code, which is written in Python.

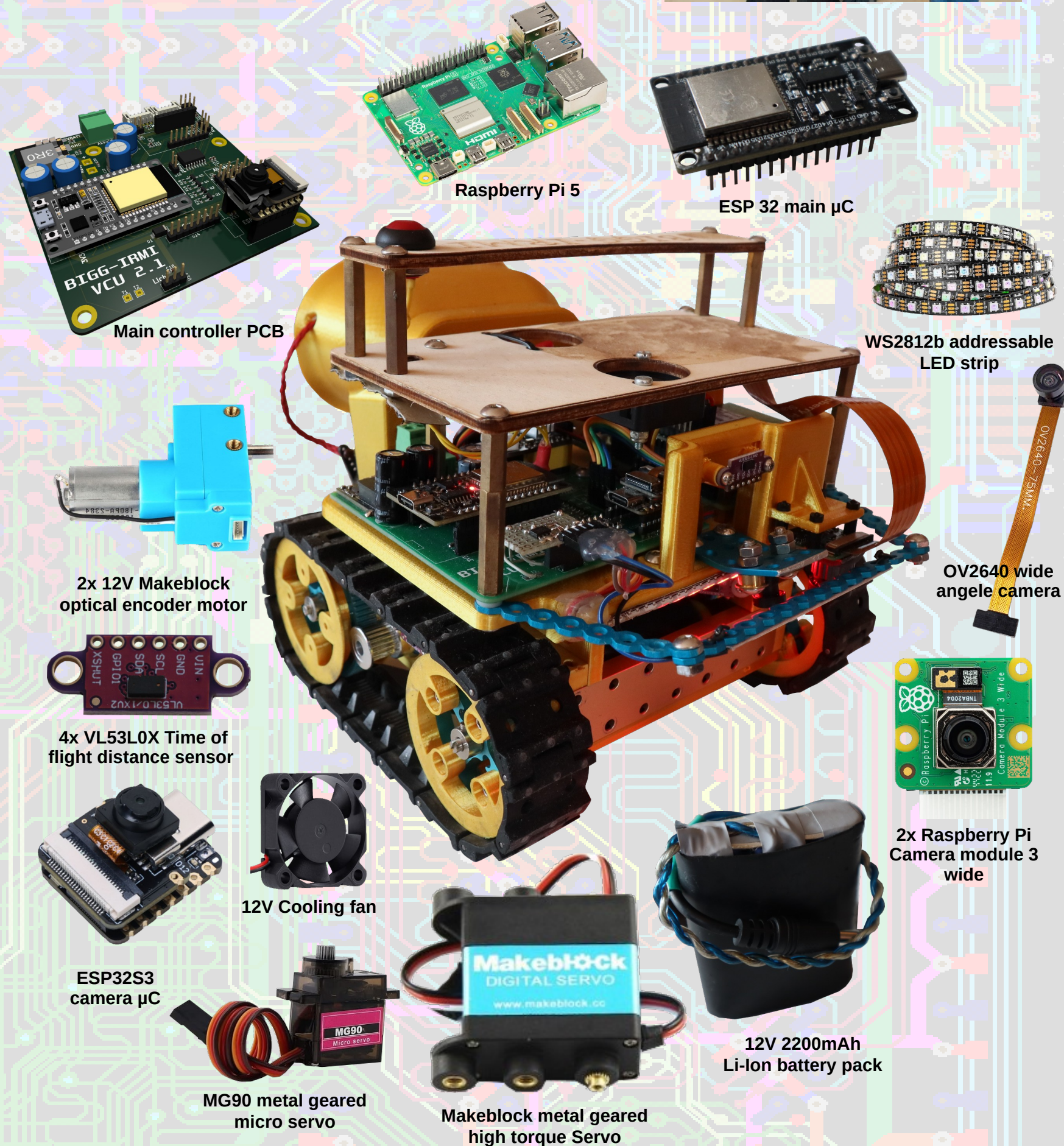
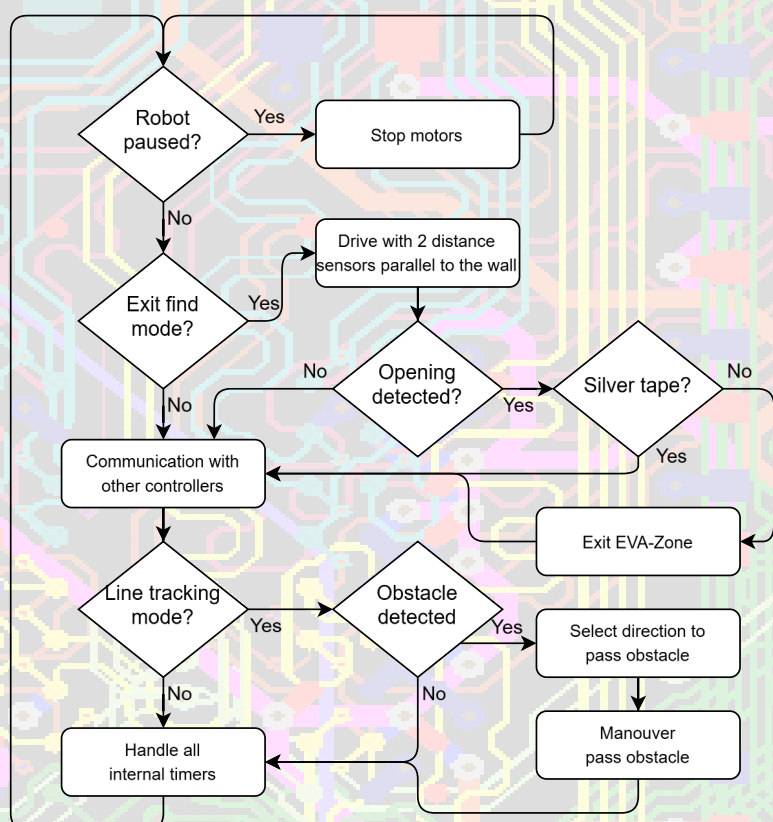


Line Tracking

The program operates using the primary front camera (ESP32S3), which contains a bespoke line tracking algorithm that has been developed from ground up. A variety of sophisticated techniques were employed to optimise runtime performance. These optimisations enable us to process approximately 20 FPS (30 ms capture, 29 ms decode, 800 µs line tracking).

Main Controller

It runs on the ESP32 main controller, which controls the motors, the victim presence sensor, the four VL53L0X distance sensors and the two servos. It also monitors the battery voltage and health of the ESP32S3 and SBC. During operation, it receives commands from the ESP32S3 or SBC and carries out tasks such as turning on the lights, measuring distances and controlling the motors. In line tracking mode, it constantly checks the front distance sensor for obstacles. It also handles start/stop commands given by the button at any time. Additionally, a watchdog is employed to ensure safety and optimal performance during operation.



Iren Biggel:

- Born in 2008, Germany
- Team captain
- Embedded software development
- Project management
- Hardware design
- Testing

Mila Biggel:

- Born in 2011, Germany
- Team member
- Mechanical CAD & CAM
- 3D manufacturing
- Performance evaluation
- Testing

Achievements

- 2026 Qualification for European Championship achieved
- 2026 3rd Place German Open Köln
- 2026 2nd Place German Open Qualification Tournament
- 2025 5th Place European Championship Bari
- 2025 5th Place German Open Nürnberg
- 2024 7th Place European Championship Hannover
- 2024 1st Place German Open Qualification Tournament
- 2022 2nd Place European Championship Portugal

Our GitHub Repository



System Layout

